

Subhadeep Ghorai

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SUMMARY

Software Engineer building AI-agent evaluation systems and full-stack platforms on the MERN stack. Shipped a VS Code extension to the Marketplace built on real Node.js runtime internals, and a merged documentation fix to **electron/electron**. Designed concurrency-safe transactions and atomic operations. Open to SDE / New Grad roles.

EXPERIENCE

Newton School of Technology

Dec 2025 – Present

Software Engineer

Remote

- Built and deployed AI-agent evaluation pipelines (**RepoScore-V2**, **WAP-Agent**) using Node.js and the Claude API, automating GitHub PR/repo grading for 441 students in ~110 minutes.
- Designed and developed internal full-stack tools (React, Node.js, Express, MongoDB) for project evaluation pipelines, learning platforms, and operational workflows used across the engineering program.
- Delivered 97 technical lectures across 9 batches on System Design, OOP, and Software Engineering fundamentals to 300+ engineers; maintained a 4.44/5 rating across 569 responses (91.55% CSAT).

Codingal

Dec 2023 – Jan 2026

Software Engineer & Senior Coding Instructor

Remote

- Designed and shipped **Atlas**, an internal operations platform (React, Node.js, Google Sheets API) replacing a manual spreadsheet/Canva workflow for a 700+ teacher organization; onboarded 100+ teachers in the first 10 days and automated 168+ certificates.
- Mentored 100+ students across 10+ countries through 2,500+ live technical sessions; promoted from Teacher Trainee to Senior Coding Instructor based on a 4.44/5 rating and 80% renewal rate.

PROJECTS

EventLoop Studio | *TypeScript, Node.js vm, Acorn, React, VS Code Extension API* [GitHub] [Marketplace]

- Built and published a VS Code extension to the VS Code Marketplace that executes JavaScript inside a real Node **vm** sandbox with native Promise intrinsics, giving spec-correct async/microtask behavior instead of simulating it from an AST.
- Engineered a scrubbable step-by-step execution replay engine capturing push/pop, console calls, and timer/microtask/phase transitions as **ExecutionStep** objects, reconstructable at any point in the trace.
- Modeled both browser (call stack, heap, Web APIs, macrotask queues) and Node.js (libuv phases + a central Microtask Hub) execution semantics for accurate cross-runtime visualization.

NSTEP – Internship Evaluation & Slot-Booking Platform | *React, Node.js, Express, MongoDB, JWT* [GitHub] [Live]

- Designed a concurrency-safe booking system for 300+ students using MongoDB transactions and atomic conditional updates to eliminate read-modify-write races; validated under load (20 simultaneous requests on one slot → exactly 5 succeed).
- Built cross-admin isolation with a custom ScheduleLock mechanism and JWT-based server-side authorization, guaranteeing zero cross-admin scheduling conflicts by design across independently managed admin sessions.
- Kept Google Sheets off the critical booking path with persisted, auto-retried async sync, and added account-based login throttling to prevent shared-campus-IP lockouts; covered by 206 unit, concurrency, and integration tests.

SESD-Agent – LLM Orchestration for Grading at Scale | *Node.js, Claude API, Google Sheets API* [GitHub]

- Orchestrated three independent evaluation pipelines (Project Evaluator, Case Study Evaluator, and a non-LLM Streak Evaluator) grading 441 student repos and 154 case studies in ~110 and ~40 minutes respectively.
- Built a GitHub/LeetCode/Codeforces contribution-calendar scorer at ~500ms per student, writing every result directly back to a shared Google Sheet used by the cohort.

OPEN SOURCE

electron/electron – Merged a documentation fix (PR #53114) closing a gap open since 2018: documented that a pending navigation measurably delays **debugger.sendCommand()** resolution (~445ms via timestamped reproduction logs) and clarified empty-result response behavior against the Electron C++ source.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java

Frontend: React, Tailwind CSS, Vite, Framer Motion

Backend: Node.js, Express.js, MongoDB, Mongoose, JWT

Cloud & DevOps: Vercel, GitHub Actions, Git

APIs & Integrations: Claude API (LLM Orchestration), GitHub API, Google Sheets API

Concepts: System Design, Data Structures & Algorithms, REST APIs, Authentication & Authorization, Concurrency Control

Problem Solving: 166+ problems solved on LeetCode

EDUCATION

VIT-AP University

B.Tech in Electronics and Communication Engineering

Expected 2026

CGPA: 8.52/10